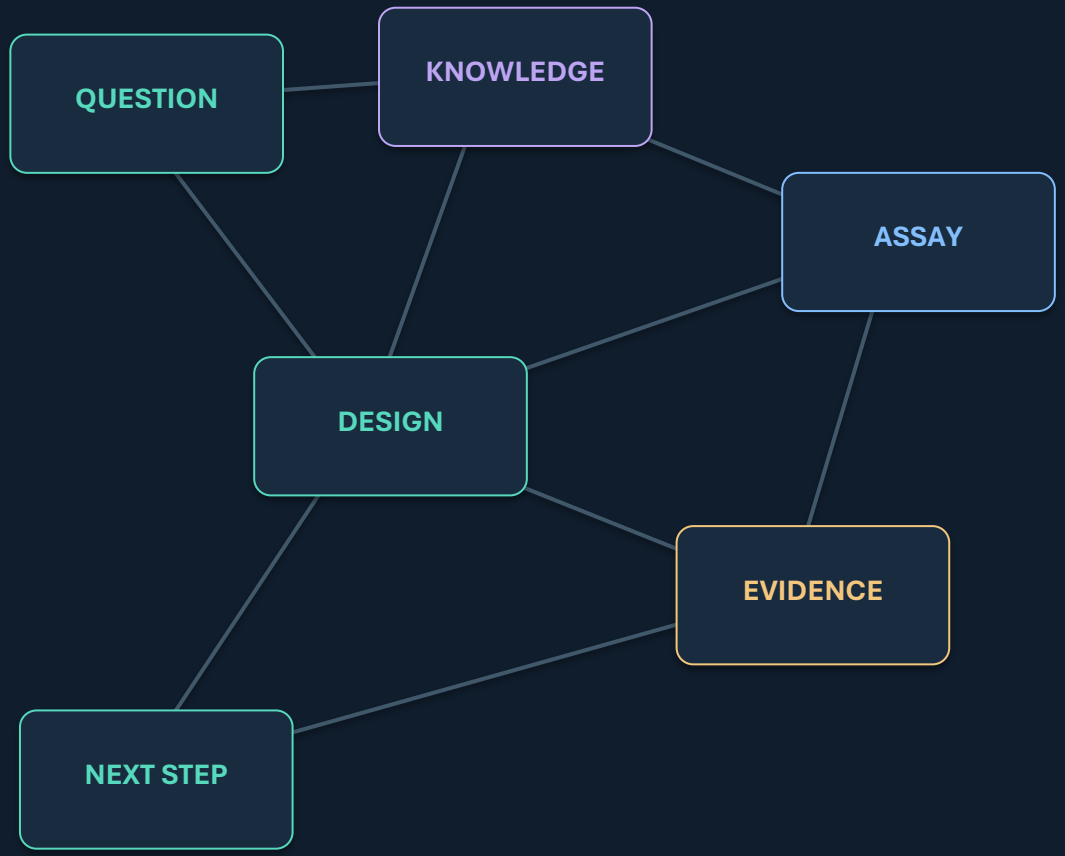
Research pathways that adapt

One engine.
Many scientific questions.

Combine materials, chemistry and biology into a research map that can change as we learn.

REUSABLE TOOLS

ADAPTIVE PATHS



Turn a research question into an evidence trail

A campaign makes the next step, the reason for it, and the result visible.

01

Define the question

What outcome matters?
What must stay fixed?

02

Choose the tools

Select relevant data, models and experiments.

03

Learn and redirect

Test a candidate.
Respond to the result.

04

Explain the decision

What did we use?
Why this next step?

The output includes the answer, the uncertainty, and the path that produced it.

Knowledge guides the work. Runs record what happened.

We author the scientific context and the research plan; execution creates the history.

DOMAIN GRAPH

What we know

Entities and mechanisms
Evidence and contradictions
Limits of our knowledge

CAMPAIGN GRAPH

What we do

Steps and decision points
Branches and repeat loops
Human approvals

DERIVED HISTORY

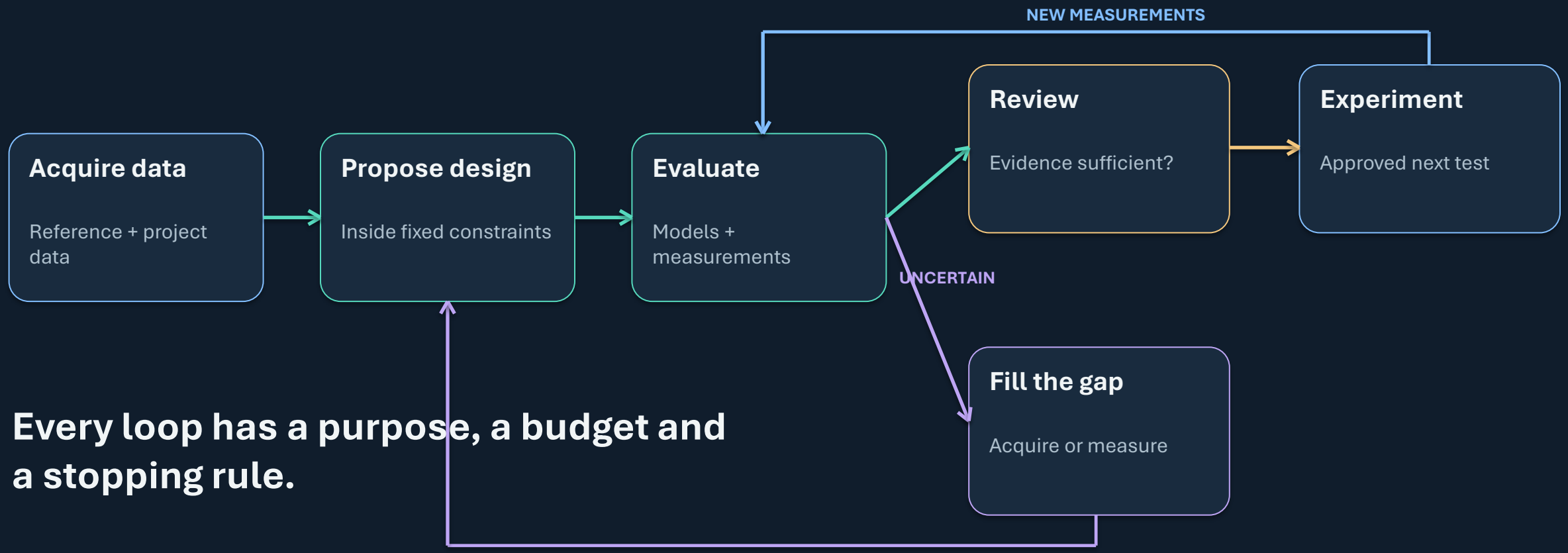
What happened

Inputs and results
Changes and failures
Who decided what

New evidence can propose updates to either map; review determines what is accepted.

The result determines where the campaign goes next

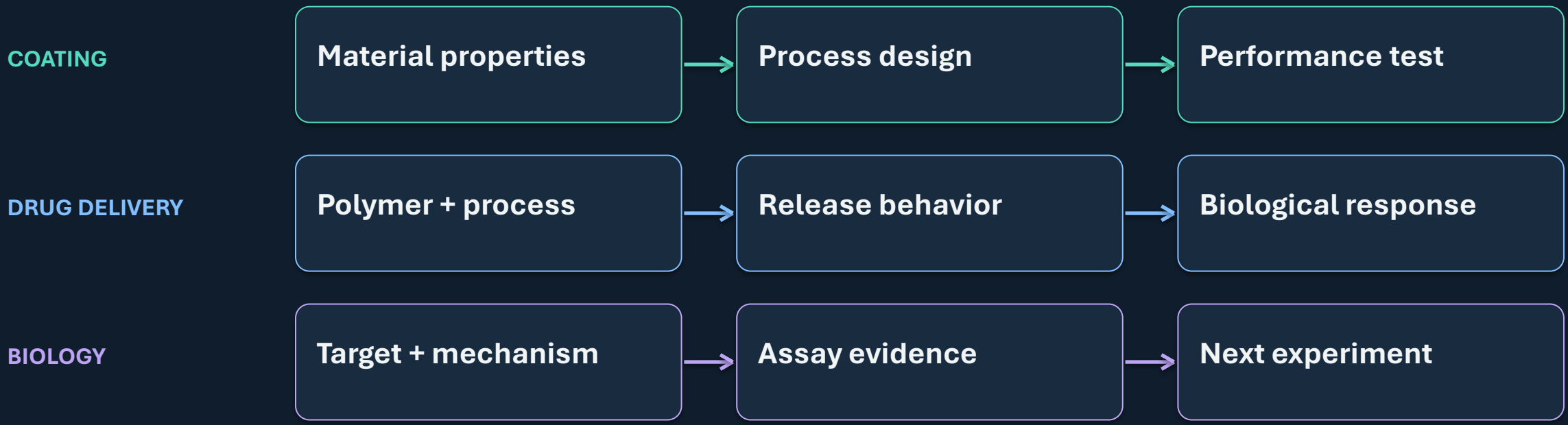
Illustrative delivery campaign: investigate, evaluate, branch, learn.



Every loop has a purpose, a budget and a stopping rule.

Use the science the question needs

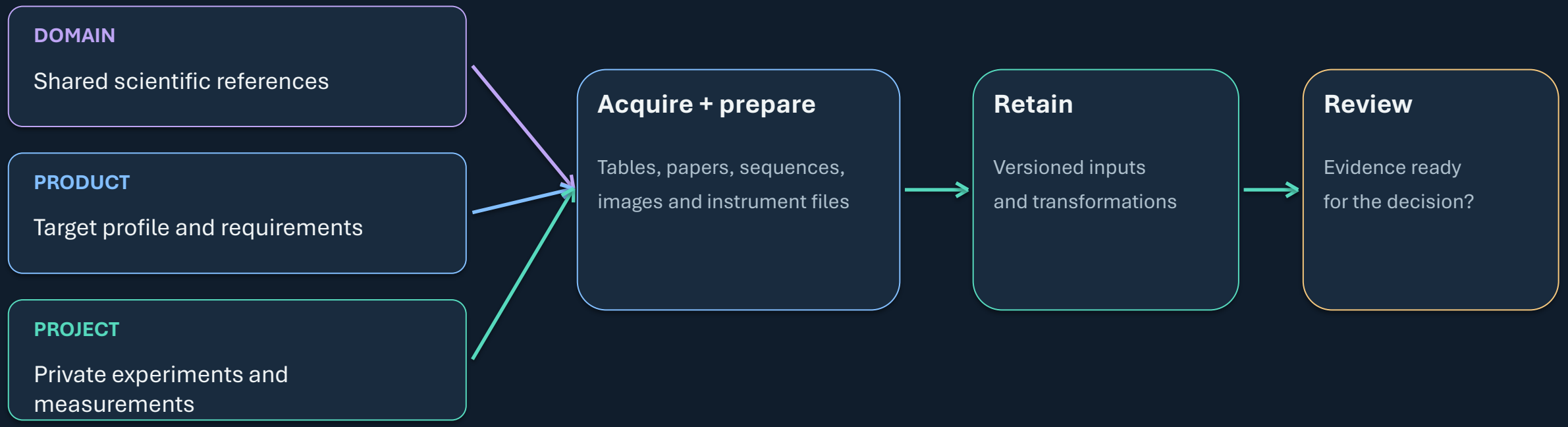
Polymers are one capability family. Biology can join the same campaign—or stand on its own.



Share capabilities when their inputs, scientific context and evidence are compatible.

Bring the right information into each campaign

Source scope controls discovery and selection; permissions still apply to every use.



A refreshed source creates a new version. Earlier decisions keep their original evidence trail.

A result has meaning only in its biological context

The engine must keep observed activity, a predicted effect and an accepted conclusion distinct.

A biological observation

What was measured?
In which system?
Under what conditions?
With what uncertainty?

SYSTEM

Species, tissue, cell line

METHOD

Assay, controls, replicates

EXPOSURE

Dose, route, time

EVIDENCE

Source, limits, contradictions

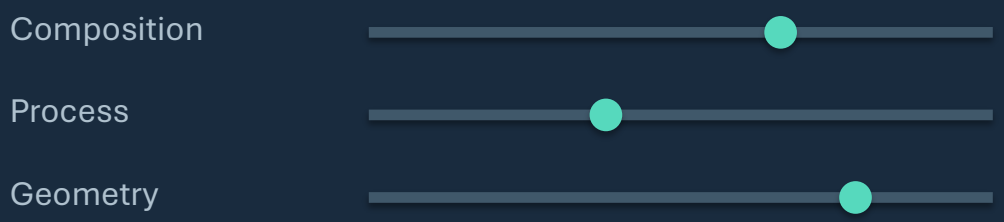
Transferring a model across products, species or routes is a new applicability decision.

Improve the candidate—and eventually the research path

These are different search spaces with different validation needs.

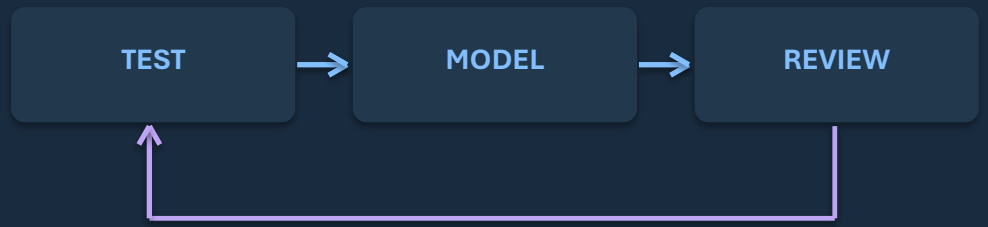
PARAMETER SEARCH / KERNEL EXISTS

Find better settings



PATHWAY EVOLUTION / FUTURE CAPABILITY

Find a better sequence of work



Evolution proposes alternatives. Constraints, evidence and human review decide what can run.

Scientists can see, question and redirect the work

Conceptual Graph Studio view: the map and the evidence remain connected.



A human intervention becomes a recorded research event—not a lost conversation.

We have reusable foundations. Scientific coverage must grow.

Source review distinguishes working algorithms from product-specific science.

REUSE

- Numerical evolutionary search
- Mixture and experiment grids
- Evidence and replay patterns

ADAPT

- Product representations
- Search-history integration
- Feature-selection workflows

BUILD + VALIDATE

- Release / biological evaluators
- Dynamic graph runtime + Studio
- Campaign-pathway evolution

Reusable optimizer code does not establish a validated drug-release or biological model.

A portfolio of research maps—not five separate engines

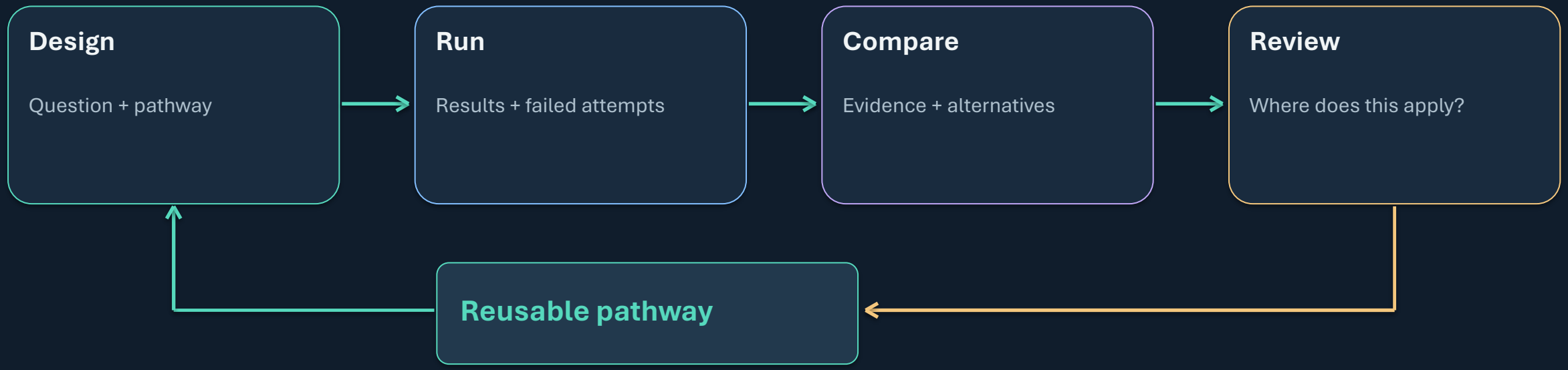
The product brief supplies examples of different scientific work on a common foundation.

PRODUCT EXAMPLE	POTENTIAL SHARED TOOLS	EVIDENCE STAYS SPECIFIC
Dexamethasone implant	Rod process + release analysis	Product, process and assay
Goserelin implant	Rod process + release analysis	Peptide stability + exposure
Once-daily buspirone	Mixture search + experiment design	Oral matrix + absorption
Veterinary analgesia depot	Depot analysis + evidence workflow	Route, species and duration
Canine OA injectable	Depot analysis + evidence workflow	Local/systemic branch + safety

Shared tools can reduce repeated engineering. Each product still has to earn its scientific evidence.

Keep the failures. Reuse the lessons. Review the transfer.

A completed campaign should improve the next research decision.



Retained history becomes useful only when context, rights and applicability travel with it.

Prove the shared engine in small, visible steps

Each gate demonstrates a new capability without claiming the next one is already complete.



GATE 1

Show portability

Rod + oral fixtures.
One search kernel.
Synthetic objectives.

GATE 2

Make it executable

Versioned graphs.
Typed tools + data.
Retained state.

GATE 3

Make it scientific

Reviewed datasets.
Validated models.
Clear assay context.

GATE 4

Make it adaptive

Human Graph Studio.
Experiment loops.
Then evolve pathways.

Biology acceptance includes a campaign that works without a polymer or epoxy node.

Build a research engine that can change its route

The scientific question determines the combination of tools.

Reusable science.
Configurable campaigns.
Visible evidence.
Human control.

REUSE THE FOUNDATION

Keep proven algorithms and evidence machinery.

EXPAND THE SCIENCE

Add biological and materials capabilities as needed.

PROVE THE CONNECTIONS

Validate the data, models and graph transitions.

First proof: the same search core, two product representations, one inspectable evidence trail.